## Introduction

In this lab, we explore the concepts of concurrency and parallelism by simulating the movement of multiple Land Mine Detector Rovers. We are to implement two approaches: a sequential method, where each rover's path is processed one after another, and a parallel method using Python's threading module to execute tasks simultaneously. The goal is to analyze the impact of parallel execution on performance and efficiency. Additionally, we need to implement a mine disarming system using brute-force hashing and compare sequential vs. threaded execution. This experiment provides valuable insights into how multithreading can optimize real-world applications involving multiple autonomous agents.

## Results

| | Sequential | Threaded |
|---|---|---|
| Part 1 | 2.21s<br>`Rover 9 commands: {"result":true,"data":{"mov`<br>`Rover 10 commands: {"result":true,"data":{"mo`<br>`Sequential Execution Time: 2.21 seconds`<br><br>`Process finished with exit code 0` | 0.53s<br>`Rover 9 commands: {"result":true,"data":{"mo`<br>`Rover 4 commands: {"result":true,"data":{"mo`<br>`Threaded Execution Time: 0.53 seconds`<br><br>`Process finished with exit code 0` |
| Part 2 | 194.19s<br>`Mine 5xbd5txxmz disarmed with PIN 10446009`<br>`Mine bhh7zc2pp1 disarmed with PIN 11113240`<br>`Sequential Disarming Time: 194.18 seconds`<br><br>`Process finished with exit code 0` | 384.50s<br>`Mine en35i7jow5 disarmed with PIN 26998126`<br>`Mine we0ufx9bpx disarmed with PIN 27313061`<br>`Threaded Disarming Time: 384.50 seconds`<br><br>`Process finished with exit code 0` |

- Part 1 threaded is 1.68 seconds faster than sequential
- Part 2 sequential is 190.31 seconds faster than threaded
- I ran these codes in the Ubuntu environment using VirtualBox. The times I have gotten here are slower compared to when I ran the codes in my Windows environment. But they are pretty much the same thing. Windows environment - Part 1 Sequential: 0.97s, Part 1 Threaded: 0.15s, Part 2 Sequential: 251s, Part 2 Threaded: 306s

## Conclusion

Through this lab, I have successfully implemented and analyzed both sequential and parallel approaches for simulating rover movement and mine disarming. My results showed that parallel execution significantly reduces processing time by allowing multiple tasks to run concurrently. This experiment highlights the practical benefits of parallel computing, particularly in scenarios involving real-time data processing, robotics, and distributed systems. Understanding when and how to apply concurrency techniques can greatly enhance system efficiency and performance in real-world applications.